Sri Lanka Institute of Information Technology

Information Technology

Partial Fractions

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Outline

1. Learning outcomes
2. Partial fractions
3. Denominators with quadratic factors

Learning outcomes

- Factorize the denominator of an algebraic fraction into its prime factors
- Separate an algebraic fraction into its partial fractions
- Recognise the rules of partial fractions

Partial fractions

To reverse the process, namely, to separate an algebraic fraction into its partial fractions we proceed as follows. Consider the fraction:

$$\frac{8x-28}{x^2-6x+8}$$

Firstly, the denominator is factorized to give:

$$\frac{8x-28}{x^2-6x+8}=\frac{8x-28}{(x-2)(x-4)}$$

Partial fractions

Next, it is assumed that a partial fraction break down is possible in the form:

$$\frac{8x-28}{(x-2)(x-4)} = \frac{A}{x-2} + \frac{B}{x-4}$$

The assumption is validated by finding the values of A and B

Partial fractions

To find the values of A and B the two partial fractions are added to give:

$$\frac{8x-28}{(x-2)(x-4)} = \frac{A}{x-2} + \frac{B}{x-4}$$

$$= \frac{A(x-4)+B(x-2)}{(x-2)(x-4)}$$

Partial fractions

Since:

$$\frac{8x-28}{(x-2)(x-4)} = \frac{A(x-4)+B(x-2)}{(x-2)(x-4)}$$

And since the denominators are identical the numerators must be identical as well. That is:

$$8x-28 \equiv A(x-4)+B(x-2)$$

Partial fractions

Consider the identity:

$$8x-28 \equiv A(x-4)+B(x-2)$$

$$\text{Let } x=4 \text{ then } 32-28 \equiv A(0)+B(2) \text{ so } B=2$$

$$\text{Let } x=2 \text{ then } 16-28 \equiv A(-2)+B(0) \text{ so } A=6$$

Therefore:

$$\frac{8x-28}{(x-2)(x-4)} = \frac{6}{x-2} + \frac{2}{x-4}$$

Partial fractions

For this procedure to be successful the numerator of the original fraction must be of at least one degree less than the degree of the denominator. If this is not the case the original fraction must be reduced by division. For example:

$$\frac{x^2+3x-10}{x^2-2x-3}=\frac{x^2-2x-3+5x-7}{x^2-2x-3}$$

$$=1+\frac{5x-7}{x^2-2x-3}$$

$$=1+\frac{5x-7}{(x+1)(x-3)}=1+\frac{3}{x+1}+\frac{2}{x-3}$$

Denominators with quadratic factors

A similar procedure is applied if one of the factors in the denominator is a quadratic. For example:

$$\frac{15x^2-x+2}{(x-5)(3x^2+4x-2)}=\frac{A}{x-5}+\frac{Bx+C}{3x^2+4x-2}$$

This results in:

$$15x^2-x+2\equiv A(3x^2+4x-2)+(Bx+C)(x-5)$$

$$=(3A+B)x^2+(4A-5B+C)x-2A-5C$$

Denominators with quadratic factors

Equating coefficients of powers of x yields:

$$[x^2] \quad 15 = 3A + B$$

$$[x] \quad -1 = 4A - 5B + C$$

$$[CT] \quad 2 = -2A - 5C$$

Three equations in three unknowns with solution:

A = 4, B = 3 and C = -2 so that:

Denominators with quadratic factors

Substituting A = 4, B = 3 and C = 2

$$\frac{15x^2-x+2}{(x-5)(3x^2+4x-2)}=\frac{4}{x-5}+\frac{3x-2}{3x^2+4x-2}$$

Denominators with quadratic factors

Repeated factors in the denominator of the original fraction of the form:

$$(ax+b)^2$$

give partial fractions of the form:

$$\frac{A}{ax+b}+\frac{B}{(ax+b)^2}$$

Denominators with quadratic factors

Similarly, repeated factors in the denominator of the original fraction of the form:

$$(ax+b)^3$$

give partial fractions of the form:

$$\frac{A}{ax+b}+\frac{B}{(ax+b)^2}+\frac{C}{(ax+b)^3}$$

Thank You!